

Locality Beats Widest-Path: DAG Scheduling under Wireless Edge Interference

Maya Gutierrez
University of Southern California
maya.gutierrez@usc.edu

Jared Coleman
Loyola Marymount University
jared.coleman@lmu.edu

Bhaskar Krishnamachari
University of Southern California
bhaskar@usc.edu

Abstract—In tactical and Internet-of-Battlefield-Things deployments, compute workloads such as sensor fusion pipelines are commonly expressed as directed acyclic graphs of tasks that must be assigned to wireless edge nodes connected by multi-hop ad-hoc links. The widely used Heterogeneous Earliest Finish Time (HEFT) scheduler and its variants were designed for settings in which the bandwidth between any pair of nodes can be treated as a fixed value. On a shared wireless channel under carrier-sense multiple access with collision avoidance (CSMA/CA), however, every additional active transmission reduces the effective rate available to all other flows in the same interference neighborhood, so the static-bandwidth assumption no longer holds. We use discrete-event simulation of an 802.11ax network, employing a validated flow-level model of CSMA/CA contention, to quantify the resulting mismatch. We examine two bandwidth matrices for HEFT. Locality-penalized HEFT (HEFT-1) applies a heavy penalty to non-adjacent placements; Widest-path HEFT (HEFT-2) uses the highest available path bandwidth between any pair of nodes. The locality-penalized variant recovers simulated makespans $2.3\text{--}10.1\times$ better than the widest-path variant across the configurations we study. When HEFT is instead supplied with widest-path estimates, its analytic predictions are $9\text{--}70\times$ optimistic relative to simulation because the model does not see the contention generated by the placements it produces. Among nine routing schemes the interference-unaware shortest-hop rule achieves the lowest makespan in the largest number of evaluated configurations.

Index Terms—edge computing, DAG scheduling, HEFT, wireless interference, CSMA/CA, routing, discrete-event simulation

I. INTRODUCTION

Tactical and Internet-of-Battlefield-Things (IoBT) deployments increasingly push compute workloads onto wireless edge devices that are connected via multi-hop ad-hoc links rather than wired infrastructure [1], [2]. Concrete examples include ISR processing pipelines in which acoustic, RF, or video sensors generate local features, intermediate nodes perform fusion or classification, and downstream tasks aggregate detections for decision support over an ad hoc wireless mesh. Many of these workloads admit a natural representation as a directed acyclic graph (DAG) of compute tasks with data dependencies, and the question of how to assign tasks to nodes and route the inter-task data flows is the classical problem of *DAG scheduling on heterogeneous platforms*. The Heterogeneous Earliest Finish Time (HEFT) algorithm of Topcuoglu et al. [3] remains the dominant list-scheduling heuristic for this setting and underpins many open implementations [4].

HEFT and its variants assume that the available bandwidth between any two nodes is a static scalar: a single number that depends only on the endpoints, not on what other transfers happen to be in flight at the same time. On wired clusters this is a serviceable approximation. On a shared wireless medium it is not: every active link consumes spectrum, and so the effective throughput of any one transfer collapses with the number of simultaneously active links in its neighborhood. Long multi-hop routes are particularly costly because each relay introduces an additional transmitter that interferes with all other concurrent flows.

To understand the practical consequences of this mismatch, we evaluate HEFT under an explicit model of wireless interference. We use *ncsim*, a discrete-event simulator of an 802.11ax wireless edge network that incorporates the closed-form Bianchi model for CSMA/CA contention [6]. Our experiments compare two bandwidth matrices for use with HEFT: a locality-penalized matrix (Locality-penalized HEFT, HEFT-1) and the conventional widest-path matrix (Widest-path HEFT, HEFT-2). Because the inter-task data transfers share wireless links and contend for spectrum under CSMA/CA, the particular routes chosen for those transfers also affect the resulting interference and intra-link sharing, and therefore the realized makespan. We evaluate the two placement strategies in combination with nine routing schemes spanning simple baselines, static greedy planners, and dynamic contention-aware approaches, across grids and random geometric graphs at multiple densities and for DAGs with 8 to 60 tasks. The key contributions of this work are:

- 1) We show that the static all-pairs bandwidth abstraction underlying HEFT can induce globally poor placements in shared wireless media. When the matrix is populated with widest-path bandwidths (Widest-path HEFT, HEFT-2), the analytic predictions are 9 to 70 times lower than the makespans measured in simulation, because the model does not account for the contention that the placement itself creates (Section V-A).
- 2) We show that a simple locality prior in the bandwidth matrix (Locality-penalized HEFT, HEFT-1) reduces the realized makespan by factors of 2.3 to 10.1 relative to HEFT-2 by driving HEFT toward co-location and adjacent placement. A sensitivity sweep confirms that this improvement holds across four orders of magnitude

in the penalty value (Section V-A).

- 3) Among the nine routing schemes, the simplest interference-unaware shortest-by-hop rule achieves the lowest mean makespan in the largest number of (network, DAG-size) configurations. Deferred dynamic routing (GSD-D) provides additional gains on sparse networks where few alternative paths exist (Section V-B).
- 4) We report auxiliary metrics, including mean hop count and peak link utilization, that clarify why particular combinations perform well (Section V-C).
- 5) We confirm through a no-interference baseline experiment (Section V-D) that the primary performance differences we observe disappear when CSMA/CA contention is removed, establishing inter-link interference as the responsible mechanism.

The remainder of this paper is organized as follows. Section II presents the network and interference model together with the two HEFT variants. Section III describes the nine routing schemes under study. Section IV details the experimental setup and simulator validation. Section V reports our simulation results and analysis. Section VI reviews related work on DAG scheduling and wireless routing, and Section VII concludes the paper.

II. SYSTEM MODEL

A. Network and DAG

A network is a set of compute nodes $\mathcal{N}$ and a set of wireless links $\mathcal{L} \subseteq \mathcal{N} \times \mathcal{N}$. Each node $n \in \mathcal{N}$ has a compute capacity C_n in compute-units per second; each link $\ell = (u, v)$ has a nominal bandwidth b_ℓ (MB/s) and propagation latency δ_ℓ (s). A workload is a DAG $G = (T, E)$: tasks $t \in T$ have compute cost w_t (compute-units) and edges $(t, t') \in E$ carry data of size $d_{t,t'}$ (MB). When a task t runs on node n its execution time is w_t/C_n . When the data $d_{t,t'}$ traverses link ℓ alone it takes $d_{t,t'}/b_\ell + \delta_\ell$ seconds.

B. Bandwidth sharing and interference

Two effects modulate the nominal link bandwidth b_ℓ at runtime. First, *intra-link fair sharing*: if k flows traverse the same link, each receives a $1/k$ share of the link's bandwidth. Second, *inter-link interference*: simultaneously active links in the same neighborhood share the shared wireless medium. The interference factor $f_\ell(\mathcal{A}) \in (0, 1]$ is computed in two steps for each active link ℓ and current active-link set $\mathcal{A}$. (i) A *conflict graph* is built from the receiver-side carrier-sensing threshold (-82 dBm CCA): two links conflict if either link's transmission would be sensed by the other's transmitter, computed from P_{tx} , the log-distance path-loss model ($P_{tx} = 20$ dBm, exponent $n = 3$), and the receiver geometry. Conflicting active links share airtime via the closed-form Bianchi DCF efficiency [5]: with n contending links each gets an $\eta(n)/n$ share of the channel. (ii) Active links *outside* ℓ 's conflict set are hidden terminals and may transmit concurrently; their aggregate received power is combined with the noise floor (-95 dBm) into an SINR computation, and the resulting rate (capped at the link's clean-channel PHY rate) is used as the effective throughput. Each

TABLE I

THE TWO HEFT BANDWIDTH-MATRIX VARIANTS COMPARED IN THIS PAPER. HEFT-1 IS LOCALITY-PENALIZED HEFT (HEAVY PENALTY ON NON-ADJACENT PAIRS); HEFT-2 IS WIDEST-PATH HEFT (WIDEST-PATH BANDWIDTH USED FOR ALL PAIRS).

Variant	Adjacent $(u, v) \in \mathcal{L}$	Non-adjacent
HEFT-2	widest-path PHY rate	widest-path PHY rate
HEFT-1	direct-link PHY rate	0.001 MB/s penalty

hidden terminal's transmission probability is weighted by its own contention domain occupancy. The product of the contention-domain share and the SINR-derived rate ratio yields $f_\ell(\mathcal{A})$. Effects *not* modeled include capture beyond the simple SINR threshold, RTS/CTS overhead (configurable but disabled in our runs), MAC retries beyond Bianchi's implicit assumption of saturated buffers, and any spatial reuse beyond what the static conflict graph captures. Composing intra-link sharing and the interference factor, the effective per-flow throughput on link ℓ is

$$b_\ell^{\text{eff}} = \frac{b_\ell f_\ell(\mathcal{A})}{k_\ell}, \quad (1)$$

where k_ℓ is the number of flows on ℓ . As flows start and finish, the simulator updates $\mathcal{A}$ and recomputes b_ℓ^{eff} for all affected links.

The interference model described above, including the computation of the interference factor $f_\ell(\mathcal{A})$ and the resulting effective bandwidth (Eq. (1)), is implemented in the `ncsim` discrete-event simulator used for all experiments in this paper.

C. HEFT and its bandwidth matrix

HEFT [3] computes a placement and an estimated finish time for each task by, at each step, placing the highest-upward-rank ready task on the node that minimizes its earliest finish time:

$$\text{EFT}(t, n) = \text{EST}(t, n) + \frac{w_t}{C_n}, \quad (2)$$

$$\text{EST}(t, n) = \max \left(a_n, \max_{p \in \text{pred}(t)} [\text{EFT}(p, n_p) + d_{p,t}/R(n_p, n)] \right), \quad (3)$$

where a_n is when node n next becomes available and $R(n_p, n)$ is the *pairwise bandwidth* HEFT assumes between nodes n_p and n . The function $R(\cdot, \cdot)$ is a static bandwidth matrix supplied to HEFT at the start of scheduling; the algorithm is unaware of any runtime contention.

The choice of $R(\cdot, \cdot)$ is a modeling decision, not part of the HEFT algorithm itself. We examine two variants, summarized in Table I.

Locality-penalized HEFT (HEFT-1) applies a heavy penalty to non-adjacent node pairs. This encodes the modeling prior that communicating tasks should almost always be placed on the same node or on directly adjacent nodes. Widest-path HEFT (HEFT-2) instead uses the highest available path bandwidth between any pair of nodes; this is the more conventional choice when one assumes that multi-hop routes perform roughly as

TABLE II

ROUTING SCHEMES COMPARED. STATIC SCHEMES PLAN ALL ROUTES ONCE AT DAG INJECTION. DYNAMIC SCHEMES RECOMPUTE THE PATH AT EVERY TRANSFER-START EVENT.

Code	Class	Objective
W	baseline	widest-path: $\max \min_{\ell} b_{\ell}$
S	baseline	shortest-path: $\min \sum_{\ell} 1/b_{\ell}$
SH	baseline	shortest-hop, tie-break $\sum_{\ell} 1/b_{\ell}$
GS	stat. greedy	order flows by start time
GC	stat. greedy	order flows by upward-rank (ranku)
GB	stat. greedy	order flows by data size (largest first)
GO	stat. greedy	order flows by total temporal overlap
GSD	dynamic	score $\min_{\ell} b_{\ell} f_{\ell} / (k_{\ell} + 1)$
GSD-D	dyn. + defer	GSD; defer if $\text{eff./uncong.} < 0.3$

well as their best individual link. We treat the penalty value in HEFT-1 as a hyperparameter and report a sensitivity sweep in Section V-A.

III. ROUTING SCHEMES

The scheduler decides *where* each task runs; a separate routing component decides *which links* carry each inter-task data flow. We compare nine schemes, listed in Table II. Three are baselines that ignore interference; four are static greedy schemes that pick paths to maximize total system throughput under HEFT’s predicted concurrency; two are dynamic schemes that recompute paths at flow start time using the simulator’s actual active-link state.

The four static greedy schemes share the same scoring function, evaluated when each candidate path is considered:

$$\text{score}(\text{path}) = \sum_{f \in \mathcal{F}_t} \min_{\ell \in \text{path}(f)} b_{\ell}^{\text{eff}}, \quad (4)$$

where $\mathcal{F}_t$ is the set of flows whose HEFT-predicted time windows overlap with the candidate’s. The four variants differ only in the order in which flows are presented to the greedy planner.

The dynamic schemes (GSD, GSD-D) recompute the path inside the `transfer_start` event handler, using the actual set of currently active links and per-link transfer counts from the simulator. GSD picks the single path that maximizes its own bottleneck throughput (Eq. 1), ignoring siblings. GSD-D extends GSD with a deferral mechanism: if the newly chosen path’s effective bottleneck bandwidth is below 30% of its uncongested value, the flow is postponed to the next `transfer_complete` event (with a starvation-prevention cap of three deferrals).

IV. EXPERIMENTAL SETUP

a) *Simulator*.: All simulations are performed using `ncsim`, an open-source discrete-event simulator that implements the network and interference model of Section II (including the CSMA/CA interference factor of Eq. (1)) with microsecond precision.¹ The flow-level CSMA/CA model is validated in the companion paper [6].

¹`ncsim` source and the scenarios used in this paper are available at <https://github.com/ANRGUSC/ncsim>.

TABLE III

HEFT-PREDICTED (ANALYTIC) VS. NCSIM-SIMULATED MAKESPAN IN SECONDS, FIXED SHORTEST-HOP (SH) ROUTING. “S” = SMALL DAG (8 TASKS), “L” = LARGE DAG (30 TASKS). PREDICTED COLUMNS ARE DETERMINISTIC; NCSIM COLUMNS SHOW MEAN $\pm$ 95% CI HALF-WIDTH OVER 30 SEEDS (OMITTED WHERE INDISTINGUISHABLE FROM ZERO).

Topology	HEFT-1		HEFT-2	
	predicted	ncsim	predicted	ncsim
4×4 S	11.0	18.3	10.6	93.9
4×4 L	4,021.9	292.4	23.9	1,071.7
7×7 S	10.9	20.3	10.2	93.9 ± 0.2
7×7 L	18,039.0	169.1 ± 7.2	24.8	$1,715.0 \pm 78.6$

b) *Topologies*.: We evaluate three classes of network. Two regular grids (4×4 and 7×7) with 40 m node spacing serve as controlled topologies. We also generate random geometric graphs of 50 nodes placed uniformly in an $L \times L$ square with communication range $R = 80$ m, varying L from 150 m (avg. degree 24.0, dense) through 500 m (avg. degree 3.7, sparse). Topology seeds are fixed per density level; only DAG / scheduler state randomization varies across replications.

c) *DAG workloads*.: For grid experiments we use a small DAG (8 tasks, fork-join) and a large DAG (30 tasks, 5-stage pipeline). For DAG-scaling experiments we vary task count from 8 to 60 (fork-join, 3-, 4-, 5-, and 6-stage pipelines). Compute costs are heterogeneous in $[150, 1000]$ cu, data sizes in $[2, 30]$ MB, node capacities in $[80, 300]$ cu/s.

d) *Replication*.: Mean makespan is reported with 95% confidence intervals over 20–30 random seeds per (topology, DAG, scheduler, routing) configuration.

V. RESULTS

In this section, we present the results of our simulation experiments comparing HEFT variants and routing schemes under the CSMA/CA interference model.

A. Scheduler choice: HEFT-1 vs. HEFT-2 under interference

Table III compares the makespan HEFT *predicts* from its bandwidth matrix against the makespan `ncsim` actually *measures* when the same placement is executed under CSMA/CA interference and shortest-hop routing. Two facts stand out. First, HEFT-2’s prediction (10–25 s) is systematically optimistic: simulation produces $9 \times$ – $70 \times$ larger makespans because HEFT-2 assumes multi-hop routes are as efficient as the widest path, ignoring that each new active link reduces the effective throughput of every other simultaneously active link. Second, HEFT-1’s prediction is, by contrast, *pessimistic* on the large DAGs (4022 s for 4×4 L; 18039 s for 7×7 L): the 0.001 MB/s non-adjacent penalty makes any multi-hop alternative look disastrous, forcing co-location and yielding what HEFT thinks is a severe compute-queuing bottleneck. In simulation, however, the compute bottleneck is a fraction of what HEFT estimates: tasks queue on a few high-capacity nodes but never compete for the wireless medium. The result is a rank inversion *between the two analytic estimates and the simulated outcome*: at schedule

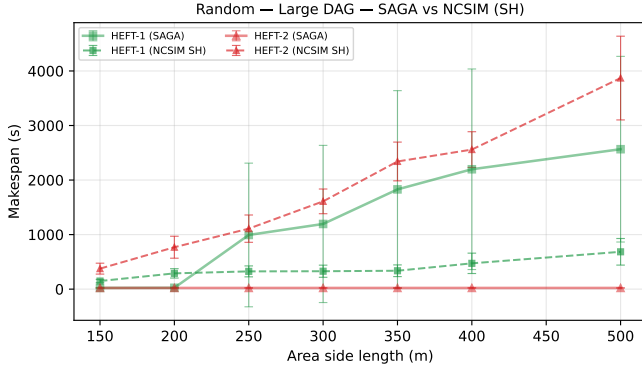


Fig. 1. Predicted (solid) vs. ncsim with SH routing (dashed) makespan as network density varies, large DAG, random geometric graphs. HEFT-2’s prediction stays low at every density; the simulated values are an order of magnitude higher and grow as the network sparsens. HEFT-1’s analytic estimate is more pessimistic but tracks the simulation more faithfully.

time, HEFT-2 appears to dominate HEFT-1 by a factor of $168\times$ – $727\times$ (the ratio of their analytic estimates on the large-DAG rows of Table III); in simulation the ranking flips and HEFT-1 wins by $3.7\times$ on $4\times 4L$ and $10.1\times$ on $7\times 7L$.

The same effect is visible across the random-network density sweep (Fig. 1): HEFT-2’s analytic estimate is essentially flat at all densities while the simulated makespan rises sharply as the graph sparsens and HEFT-2’s preferred multi-hop routes lengthen.

a) *Mechanism.*: HEFT-2 produces well load-balanced placements because widest-path bandwidth makes “put this task on a far node” look cheap; the resulting placement has many cross-node edges and thus many simultaneously active wireless links. Each of those links degrades every other link’s effective bandwidth, and the global makespan is driven by the slowest of the resulting flows. HEFT-1’s heavy non-adjacent penalty removes the “put it far away” option from the placement search, eliminating most cross-node data movement and the interference it would have generated.

b) *Penalty sensitivity.*: HEFT-1’s effect depends only weakly on the exact value of the non-adjacent penalty rate: any value small enough that HEFT prefers a same-node or one-hop placement over a multi-hop alternative produces essentially the same placement. Fig. 2 sweeps the penalty rate over four orders of magnitude (10^{-4} to 10^{-1} MB/s) on two grid configurations and three random densities, with SH routing and 30 seeds per cell. Across this entire range the simulated makespan is essentially flat (within seed noise: e.g., 146–164 s on L150, 292 s on 4×4 , 167–171 s on 7×7). Only when the penalty is raised to 1.0 MB/s (approaching the actual link bandwidth, so the “penalty” is no longer a penalty) does HEFT begin to treat non-adjacent placements as competitive again, and the simulated makespan jumps sharply (e.g., $292\rightarrow 505$ s on the 4×4 grid; $167\rightarrow 972$ s on 7×7 , where path diversity makes the recovered HEFT-2-style placement particularly costly). The key result for HEFT-1 therefore does not depend on the specific value 0.001 MB/s used in the rest of this paper; any value at

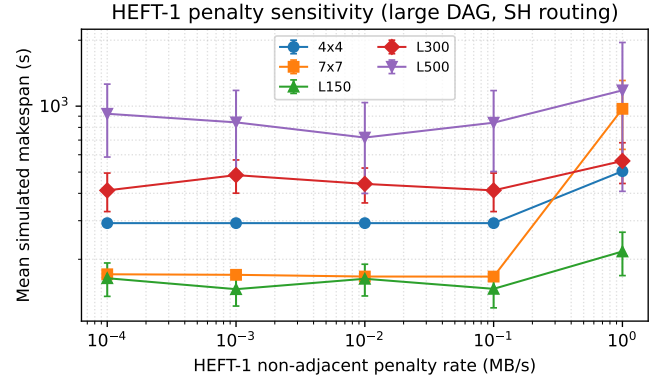


Fig. 2. Sensitivity of HEFT-1’s simulated makespan (large DAG, SH routing, 30 seeds, csma_bianchi interference) to the non-adjacent bandwidth penalty rate, swept logarithmically. The placement, and hence the makespan, is invariant across four orders of magnitude (10^{-4} to 10^{-1} MB/s); only when the penalty is raised to 1.0 MB/s does HEFT begin to choose multi-hop placements again, producing the sharp rightmost rise.

TABLE IV
NUMBER OF (NETWORK, DAG-SIZE) CELLS IN WHICH EACH ROUTING SCHEME ACHIEVES THE LOWEST MEAN MAKESPAN, OUT OF 18 CELLS TOTAL (3 NETWORKS $\times$ 6 DAG SIZES). HEFT-1 FIXED.

Routing	Total	L150	L500	7×7
SH	6	5	0	1
S	5	1	1	3
GSD-D	4	0	3	1
GB	2	0	1	1
GS	1	0	1	0
W, GC, GO, GSD	0	0	0	0

least an order of magnitude below the link bandwidth recovers the same placement.

B. Routing scheme comparison

Having fixed HEFT-1 as the placement strategy, we now ask which routing scheme to pair it with. We sweep DAG size from 8 to 60 tasks on three representative networks (L150 dense, L500 sparse, 7×7 grid) and report mean makespan over 20 seeds for each of the nine schemes. Per-network curves are shown in Fig. 3; aggregate win counts across all 18 (network, DAG-size) cells appear in Table IV.

Three observations follow.

a) *Dense networks: SH dominates and interference-aware greedy degrades by an order of magnitude.*: On L150, SH wins at every DAG size from 16 tasks onward (S edges out SH at 8 tasks). The four static greedy schemes are an order of magnitude worse: 4,500–5,000 s at 60 tasks against SH’s ~ 412 s. The mechanism is a double-counting failure of the prediction step: greedy planners score candidate paths against HEFT’s predicted concurrency window for each flow, but those windows were themselves computed under HEFT-1’s locality-biased cost model that already assumed minimal cross-link contention. The planner therefore picks paths that look interference-light against a baseline that already assumed interference was negligible,

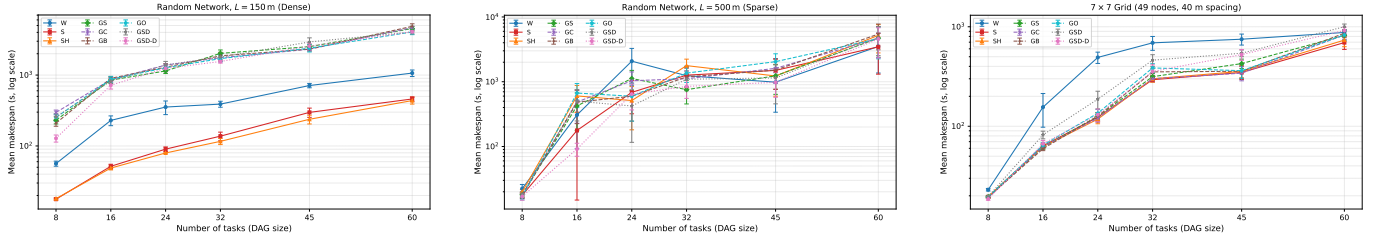


Fig. 3. Mean makespan (log scale) vs. DAG size for nine routing schemes, HEFT-1 scheduler, csma_bianchi interference. Left: dense random L150 (50 nodes, avg. degree 24.0). Middle: sparse random L500 (50 nodes, avg. degree 3.7). Right: regular 7×7 grid. Means over 20 seeds.

and at runtime the chosen paths trigger cascading recalculations that amplify the actual interference. With average degree 24 the planner has many alternatives and the cascade is severe; SH avoids the trap by minimizing the number of relay hops, which is the dominant determinant of how many simultaneously active links the schedule produces. SH is also the simplest to implement, requiring only a per-flow BFS rather than the per-overlap-window throughput optimization of GS/GC/GB/GO or the per-event path recomputation of GSD/GSD-D.

b) Sparse networks: deferral wins.: On L500 (avg. degree 3.7), GSD-D wins three of six DAG sizes. With so few path alternatives, the choice of *path* is largely determined; the lever GSD-D pulls is *when* each flow runs. By postponing a flow whose effective throughput is severely degraded by current contention, GSD-D avoids double-booking the few bottleneck links and the additional makespan cost from the deferral itself is paid back several times over. Variance is high on L500 at the largest DAG sizes ($\sigma > \mu$ at 45–60 tasks). The cause is bimodal: with average degree 3.7, many seeds yield a topology in which a single “bridge” link connects two otherwise weakly-coupled clusters, and whether HEFT happens to place a communicating task pair on opposite sides of that bridge determines whether the entire DAG’s cross-cluster transfers serialize through one wireless hop. Seeds that avoid the bridge complete quickly; seeds that don’t pay an order-of-magnitude penalty.

c) Regular grid: shortest-path leads.: On the 7×7 grid, S leads at small to medium DAG sizes; SH and S are essentially tied on large DAGs. The grid has uniform link bandwidths, so the secondary $\sum 1/b_\ell$ tiebreaker that distinguishes S and SH rarely matters; what distinguishes both from the greedy schemes is again that they minimize the number of active links rather than try to predict interference. GSD (dynamic, no deferral) is the worst scheme on the grid, behind even the static greedy variants: its repeated per-flow path recomputation produces decision-unstable choices that do not reliably improve the realized concurrency pattern.

d) W is uniformly worst.: Widest-path routing wins zero of the 18 cells. Maximizing the bottleneck bandwidth pushes every flow toward the same handful of high-capacity links, inflating both the hop count of each route and the contention on those links. Both effects compound under CSMA/CA.

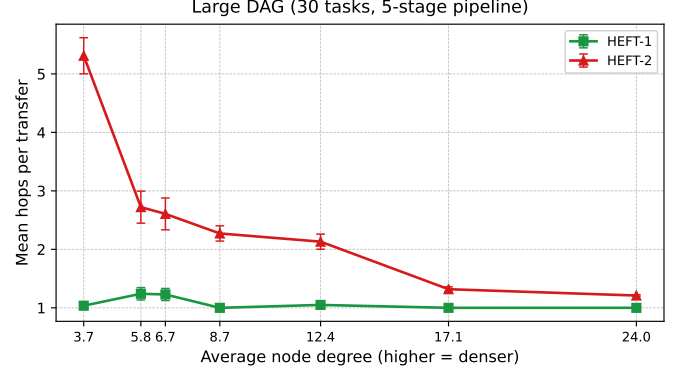


Fig. 4. Mean number of hops per inter-task transfer, large DAG, best routing scheme per (density, scheduler) cell. HEFT-1 stays at one hop across the entire density sweep; HEFT-2 grows steeply as the graph sparsens and adjacent placement becomes less feasible.

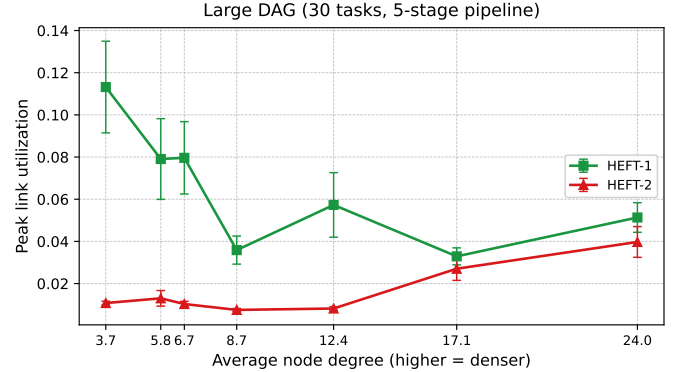


Fig. 5. Peak link utilization, large DAG, same configurations as Fig. 4. HEFT-1 keeps peak utilization low because most data transfers are eliminated by co-location; each link therefore carries little traffic.

C. Auxiliary metrics: explaining why each combination wins

Table V and Figs. 4–5 make the mechanism quantitative. Across all densities, HEFT-1 drives the mean hop count to one; essentially every transfer either vanishes into a co-located edge (zero hops) or traverses a single direct wireless link. HEFT-2, which considers any pair of nodes as roughly bandwidth-equivalent, ends up routing flows over 1.2–5.3 hops on average. The peak link utilization tells the same story from a different angle: PLU is *lower* under HEFT-1 only because the schedule

TABLE V

AUXILIARY METRICS FOR THE LARGE DAG (30 TASKS) ON REPRESENTATIVE RANDOM GEOMETRIC GRAPHS (50 NODES), AT THE BEST ROUTING SCHEME PER (DENSITY, SCHEDULER) CELL. “HOPS” IS THE MEAN HOP COUNT PER INTER-TASK TRANSFER; “PLU” IS THE PEAK LINK UTILIZATION; “MAKESPAN” IS IN SECONDS (MEAN $\pm$ 95% CI HALF-WIDTH OVER 30 SEEDS).

Network	Sched.	Routing	Hops	PLU	Makespan (s)
L150 (dense)	HEFT-1	SH	1.0	0.05	145.5 $\pm$ 23.8
L150 (dense)	HEFT-2	SH	1.2	0.04	331.6 $\pm$ 36.0
L300 (med.)	HEFT-1	SH	1.0	0.04	482.4 $\pm$ 73.1
L300 (med.)	HEFT-2	SH	2.2	0.01	2,264.8 $\pm$ 124.4
L500 (sparse)	HEFT-1	GSD-D	1.3	0.09	764.8 $\pm$ 312.6
L500 (sparse)	HEFT-2	GSD-D	5.3	0.01	5,546.6 $\pm$ 928.8

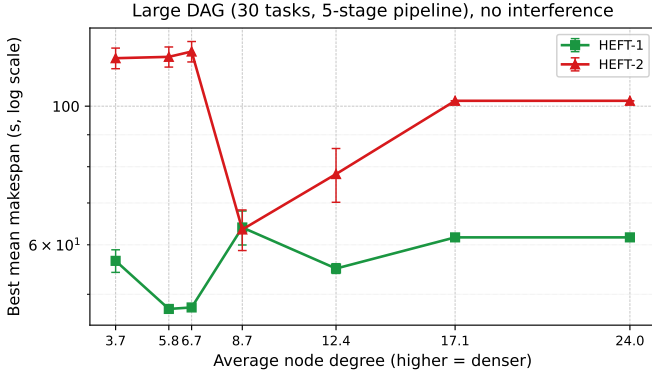


Fig. 6. Best-routing makespan vs. avg. degree, large DAG, with the inter-link interference model disabled (intra-link fair sharing only). Error bars are 95% CIs over 30 seeds. The HEFT-1/HEFT-2 simulated-makespan gap collapses from $\sim 100\times$ (interference case) to roughly $2\times$, confirming that CSMA/CA contention is the mechanism behind the primary effects. (Note that the abstract’s $9\times\text{--}70\times$ figures compare HEFT-2’s analytic predictions against its own simulated outcomes, a distinct comparison from the scheduler-to-scheduler gap discussed here.)

simply does not produce that many concurrent flows. Under HEFT-2 the few links that do see traffic see a lot of it, but the total number of active links is so much higher that the aggregate interference is severe even though no individual link is heavily loaded by its own flow alone. In other words, the relevant quantity is not “how busy is the busiest link” but “how many links are simultaneously busy.” Beyond makespan, the reduction in total wireless transmissions under HEFT-1, achieved by collapsing multi-hop relays into single-hop or co-located edges, implies a corresponding, though here unmeasured, gain in aggregate radio energy expenditure, which is directly relevant to battery-constrained IoBT and tactical edge nodes.

D. No-interference baseline

To verify that the effects above are caused by inter-link interference rather than by raw bandwidth or hop count, we re-ran the random-network density sweep with the interference model disabled (i.e., with $f_\ell \equiv 1$ in Eq. 1 but with intra-link fair sharing preserved). Fig. 6 shows the result. With interference off, HEFT-1’s makespan stays in the 47–62 s range

Comm/comp ratio sweep (large DAG, SH routing)

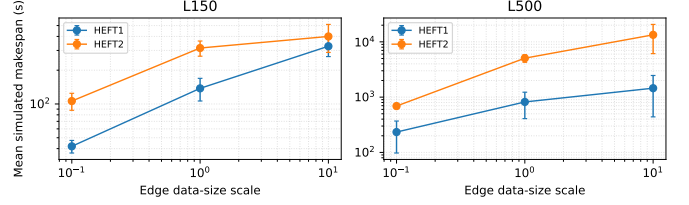


Fig. 7. Effect of scaling all DAG edge data sizes by a global factor $\{0.1, 1, 10\}$ on simulated makespan, large DAG, SH routing, 20 seeds, csma_bianchi interference. Left: dense L150; right: sparse L500. The HEFT-1/HEFT-2 gap widens with the comm/comp ratio because HEFT-2’s placement carries more bytes over the multi-hop relays that interfere most.

across all densities and HEFT-2’s is in the 70–145 s range, a $\sim 2\times$ gap rather than the $\sim 100\times$ HEFT-1/HEFT-2 gap of the interference case. The routing-scheme spread shrinks similarly. We conclude that inter-link interference is responsible for the main effects reported in this paper, not any bandwidth-modeling or hop-count artifact.

E. Communication/computation ratio

A natural concern is that the primary performance gap between HEFT-1 and HEFT-2 depends on how communication-heavy the workload is. Fig. 7 scales all DAG edge data sizes by $\{0.1, 1.0, 10.0\}$ on a dense (L150) and a sparse (L500) random network. On the sparse network the gap grows sharply with the scale factor: HEFT-2/HEFT-1 ratios of $2.9\times$, $6.2\times$, and $9.2\times$ at the three scales, with absolute makespans of 13,391 s for HEFT-2 versus 1,451 s for HEFT-1 at $10\times$. On the dense network the gap is consistently smaller ($2.5\times$, $2.3\times$, $1.2\times$), since L150’s high path diversity limits how much damage HEFT-2’s spread placement can do even at high comm/comp ratios. The mechanism is unchanged; more bytes flowing over interfering multi-hop paths costs more, but the sensitivity confirms that the locality prior is most valuable on sparse, communication-heavy workloads, the regime where IoBT and tactical edge deployments typically live.

VI. RELATED WORK

HEFT [3] and its many variants form the dominant family of list-scheduling heuristics for DAG scheduling on heterogeneous platforms. CPOP [3] prioritizes the critical path onto a single processor chosen to minimize its length. PEFT [7] uses an Optimistic Cost Table (OCT) to improve processor selection with limited lookahead during ranking. Lookahead variants of HEFT [8] consider the impact of a placement decision on immediate successors. Duplication-based extensions [9] replicate selected predecessor tasks to reduce communication delays when bandwidth is scarce. The SAGA framework [4] provides reference implementations for benchmarking many of these heuristics. However, all of these works adopt a static pairwise-bandwidth matrix that does not capture runtime interference, so the placements they produce are tuned to a cost model that does not exist in a real wireless edge.

Dependent-task (DAG) offloading and scheduling in mobile edge computing (MEC) and wireless edge environments has received substantial attention. Surveys such as Asghar et al. [10] catalog heuristic, meta-heuristic, and learning-based approaches for workflow scheduling across edge-cloud tiers, highlighting the prevalence of DAG-structured microservices. Recent learning-based methods explicitly model task dependencies via graph attention networks and deep reinforcement learning [11]. These works focus primarily on computation offloading decisions and resource allocation; they typically abstract the wireless medium as a set of time-varying but non-interfering links rather than modeling CSMA/CA contention and hidden-terminal effects at the flow level.

The wireless networking community has long developed routing metrics that account for loss, bandwidth, and interference in ad hoc and mesh networks. The ETX metric [12] estimates the expected number of transmissions (including retransmissions) required to deliver a packet, outperforming hop count on lossy links. ETT and WCETT [13] extend ETX to incorporate link bandwidth and penalize intra-flow channel reuse in multi-radio settings. The airtime metric standardized in IEEE 802.11s HWMP further refines these ideas by estimating channel occupancy time. Explicit interference-aware metrics (e.g., iAWARE, MIC) and conflict-graph formulations have been used for routing and link scheduling to mitigate inter-flow contention. However, these metrics are typically applied to unicast flows or network-layer objectives; they have not been systematically composed with DAG placement heuristics such as HEFT under a joint placement-and-routing objective.

Tactical and IoBT edge computing scenarios [1] motivate the present work: compute must be pushed onto bandwidth-constrained, contention-based wireless meshes where both placement and routing decisions affect end-to-end latency under interference. While prior work has studied DAG scheduling, wireless routing, and edge offloading separately, the interaction between DAG placement and routing under contention-based wireless interference has, to our knowledge, not been systematically evaluated for the HEFT family on representative wireless edge topologies and DAG sizes. The ncsim simulator and the present evaluation are intended to fill that specific gap.

VII. CONCLUSION

In this paper, we have shown that the cost-model abstraction used by HEFT is the source of its failure under wireless interference. When the static all-pairs bandwidth matrix is populated with widest-path estimates, the placements it produces generate many simultaneously active links that interfere with one another under CSMA/CA; the analytic model does not anticipate the resulting contention. A simple locality prior on the same matrix (a heavy bandwidth penalty on non-adjacent task pairs, HEFT-1) restores most of the lost performance by encouraging single-hop placement, and the result is robust to the exact penalty value across four orders of magnitude. Among the nine routing schemes we evaluated, the simplest interference-unaware shortest-by-hop rule wins the largest

number of configurations; deferred dynamic routing (GSD-D) provides additional gains on sparse networks where there are no good alternative paths and the only useful lever is timing. In the tested settings, shortest-hop routing is more robust than the particular greedy interference-aware routing schemes evaluated here. The likely reason is that those greedy schemes rely on predicted concurrency windows derived from an imperfect static scheduler model; thus, minimizing active hop count proves a more reliable heuristic than optimizing predicted path interference under stale concurrency estimates.

The mechanism behind these findings is specific to contention-based wireless MAC layers: under CSMA/CA, every additional active link competes for the shared channel, so multi-hop routes are severely penalized. Under scheduled MAC layers such as TDMA or 5G NR, where time-frequency resources are sliced rather than contended, the multi-hop penalty is bounded by the schedule rather than amplified by it, and load-balanced placements like HEFT-2 would be expected to recover. The prescription to favor single-hop and co-located placement therefore applies most directly to contention-based omnidirectional wireless networks such as 802.11 mesh settings. In scheduled MACs (e.g., TDMA), directional antennas, or beamformed links, the tradeoff can shift back toward load-balanced placement because the multi-hop penalty is bounded by the schedule rather than amplified by contention. Mapping the boundary between these regimes remains an important direction for future work.

In future work, we plan to extend the evaluation in four directions. First, we will model mobility, where the link set evolves during DAG execution and routing decisions must adapt online. Second, we will consider multi-DAG workloads in which placement and routing decisions must trade off across concurrent jobs. Third, we will compare the implicit co-location bias of HEFT-1 against placement heuristics that model interference explicitly inside the EFT computation. Fourth, the prescription of single-hop placement is itself a consequence of the omnidirectional shared channel; physical-layer changes such as directional antennas, beamforming, or dedicated control/data channels would shrink the interference footprint of each active link and could restore the viability of load-balanced multi-hop placements like HEFT-2. Whether the right long-term answer is hardware that makes HEFT-2 viable or software that accepts the single-hop constraint is a question we leave open.

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